

Sewershed: what we cover, what we save, and why not Boston.

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All sources verified or
labeled as estimates

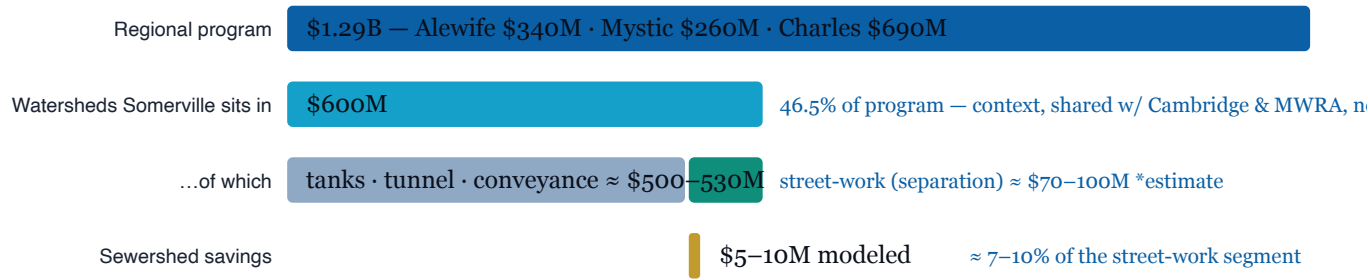
1 · What we're building and why it matters

Sewershed is a sequencing and bundling tool for street excavation. Somerville's combined sewers overflow into the Alewife Brook and Mystic River when it rains; the regional fix — the **\$1.29 billion** Cambridge + Somerville + MWRA CSO Control Plan — builds storage tanks, tunnels, and **~103 acres of targeted sewer separation in Somerville's catchments** over 13–33 years. The plan decides *what* gets built per watershed. Nothing decides *which streets open first, or what else rides in each trench*.

Sewershed joins the city's own pipe GIS (2,438 segments still labeled "Combined" — 55% of the sanitary network) with Cyvl's street-level scan (pavement condition every 30 ft, utility-cut patching on 46.5% of segments, 748 ADA ramps, sidewalk condition) and ranks every dig candidate by readiness: dig where pavement is already failing, avoid fresh asphalt, bundle the ramp and sidewalk work into the open trench. Somerville already proved the value by hand — the just-finished Spring Hill separation rebuilt streets with bike lanes and raised crosswalks while they were open. We make that systematic.

2 · The money, honestly scoped

Most of the budget is not streets — it's concrete. The savings claim must be measured against the segment we actually touch:



Scope of claim: the ~103 committed separation acres (≈21 street-miles). The same engine also covers all 2,438 combined segments, conveyance routes, water-main work, and the city's rolling dig calendar — value shown in the demo, but not claimed in dollars.

The waterfall: claim small, cover big. Bars 1–2 are from the plan's own Table 1; bar 3's split is our estimate (see note).

Figure	Value	Basis
Regional program total	\$1.29B	Plan Table 1: \$340M + \$260M + \$690M — verified
The two watersheds Somerville sits in	\$600M (46.5%)	Alewife + Mystic basin totals — shared across Cambridge, Somerville, MWRA (Alewife is heavily Cambridge-side); a context bar, not Somerville's bill — verified arithmetic
Committed separation, Somerville catchments	103 acres ≈ 21 mi	Table 1 (95 Mystic on SOM007A — solidly Somerville; 8 Alewife — possibly partly Cambridge-side, as Table 9-22 carries separation for both SOM001A and CAM001). Worst case Somerville-only: 95 acres ≈ 19.6 mi → savings ~\$4.5–9M; the claim is robust to the ambiguity — verified + geometry, attribution caveat stated
Street-work segment cost	≈ \$70–100M	Our estimate — the plan doesn't publish the split; derived from standard unit costs (large CSO tanks ≈ \$20–30M/MG, microtunnel ≈ \$100M+/mile, urban separation ≈ \$3–6M/mile)
Sewershed modeled savings	\$5–10M (\$4.5–9M worst-case attribution)	Avoided double-paving + bundled restoration + ADA-in-trench + pavement-life protection (utility cuts cut arterial life 25→16.5 yrs, LA; –4.5 yrs, Phoenix)
Savings as % of segment we touch	≈ 7–10%	\$5–10M ÷ \$70–100M street-work segment

The honest sentence (lead with bundling): "Roughly 85% of the watershed budget buys tanks and tunnels — software can't save money on concrete. The street-work segment is on the order of \$70–100M, and our \$5–10M model is about 7–10% of the money we actually touch. The most defensible piece is bundling: the ADA ramp and failing sidewalk that ride in a trench that's open anyway. Scale anchors: \$5–10M ≈ 1–1.5 years of Somerville's full paving program (~\$6.5M/yr, FY24), or \$60–120 per resident."

If every plan partner had Somerville-grade data, the same per-mile logic across all 549 separation acres (~113 street-miles) models to **\$25–55M program-wide** — stated as a projection, never a claim.

3 · Why we can't build this for the whole City of Boston (we checked — five ways)

The product needs two datasets to coexist: the city's labeled pipe network and a full street-condition scan. Boston is missing both halves, from two different owners:

Source checked (2026-06-12)	What it contains	Verdict
Cyvl "Boston Full" project (queried via API)	883 mi pavement scores + 236,983 searchable images; asset inventory empty (confirmed by direct query)	No assets → no bundling inputs
BWSC Map Room	Catch-basin/hydrant lookup viewers + CSO <i>notification</i> tools	View-only; no pipe network
data.boston.gov (catalog API searched)	"Sewer" returns 3 results: rain gauges, hydrants, a zoning overlay	No mains, no network
Boston Managed Streets (MassGIS)	Street centerlines with jurisdiction attributes	The road map, not the pipe map
Boston's own separation activity	BWSC is actively separating (South Boston project under construction; East Boston program), and Charles-watershed separation work in the regional plan reaches Boston territory	Strong future customer — but its pipe network isn't public and Cyvl didn't run asset detection there, so we can't demo it

The key nuance: the missing pipe data blocks *our demo*, not *Boston's deployment*. BWSC holds its own pipe network internally. The tool runs for Boston the day they load their own GIS — a file transfer, not a research project.

Somerville is currently the only city where both halves exist: the only municipality we found that publishes its full labeled sewer network, and the city where Cyvl ran its complete asset pipeline. That's why the demo is here — and it's a moat, not an excuse.

4 · What the numbers would be if Boston's data existed

Scenario	Modeled value	Basis
Boston's ongoing sewer capital (BWSC)	\$2–5M / year, recurring	BWSC's own 2025–27 CIP document: \$135.8M sewer capital (≈\$45M/yr) incl. active separation (South Boston live); bundling the street-surface share at our per-mile rates
...over a decade	\$20–50M	Same logic, perpetual program — a subscription, not a project
Full \$1.29B plan coverage (all partners' data)	\$25–55M	549 acres ≈ 113 street-miles at the Somerville per-mile model
Avoided inventory costs (Boston)	\$1–2M / survey cycle	~60k assets citywide at consultant field rates; Somerville recruited volunteer surveyors for its citywide ADA assessment (city announcement) — the price point in action

One-line close: "We claim \$5–10M on the 103 acres we can verify — about 7–10% of the street-work we touch. The same engine, fed the data Boston already holds internally, models to \$2–5M a year, forever. We built where the data exists; the savings scale wherever it appears next."